Chapter 6: Electronic Structure of Atoms

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CHM111

Lecture 6

Part I

Electromagnetic Energy

Outline of Part I: Electromagnetic Energy

Outline of Part I: Electromagnetic Energy

Central Motivation

Why do elements emit only specific colors of light when energized?

Spectroscopy

- The Crab Nebula (supernova remnant) glows in distinct colors.
- Green filaments: S^{+} ions   Red filaments: O^{2+} ions
- When an atom or ion loses an electron, it emits a unique "fingerprint" of light.
- Scientists use these fingerprints to learn about an observed system.

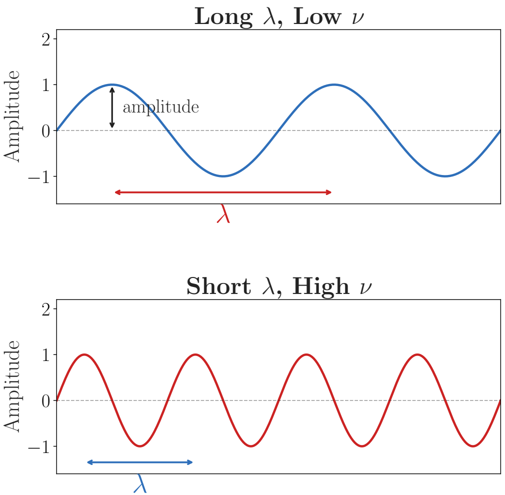
🔬 Understanding quantum theory was necessary for scientists to fully comprehend why ionized particles (atoms or ions losing electron(s)) emit light at specific wavelengths.

Outline of Part I: Electromagnetic Energy

Waves and Wave Properties

Key Wave Quantities

- **Wavelength** (λ):
  distance between consecutive peaks.
  SI Unit: m
- **Frequency** (ν):
  number of wave cycles per second.
  SI Unit: Hz $=$ s^{-1}
- **Amplitude**:
  height of the wave; related to **intensity** (brightness)
- **Speed**:
  distance traveled per unit time



The Speed of Light

Fundamental Relationship

All electromagnetic waves travel at the **speed of light** c in a vacuum:

$$\boxed{c = \lambda\nu}$$

$c = 2.998 \times 10^8\ \frac{\text{m}}{\text{s}}$ (often rounded to 3.00×10^8 m/s)

Example: Sodium Streetlight

Yellow light from a sodium lamp has $\lambda = 589\,\text{nm}$. Find ν.

$$\nu = \frac{c}{\lambda} = \frac{2.998 \times 10^8\ \frac{\text{m}}{\text{s}}}{589 \times 10^{-9}\ \text{m}}$$

$$\boxed{\nu = 5.09 \times 10^{14}\ \text{Hz}}$$

Check Your Understanding

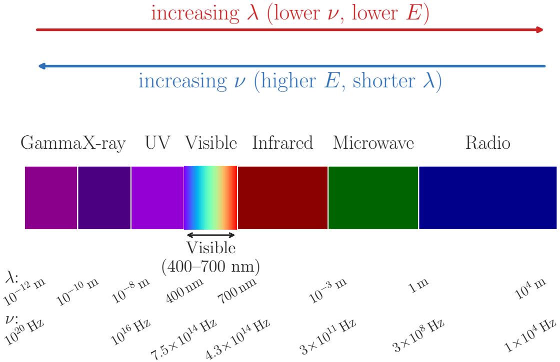
FM radio waves broadcast at 101.1 MHz ($= 1.011 \times 10^8$ Hz).

What is their wavelength in meters?

Are radio waves longer or shorter than visible light (~400–700 nm)?

Come to lecture to see the answer.

The Electromagnetic Spectrum



Spectrum Regions

Region	λ range
Gamma ray	< 0.01 nm
X-ray	0.01–10 nm
UV	10–400 nm
Visible	400–700 nm
IR	700 nm–1 mm
Microwave	1 mm–1 m
Radio	> 1 m

All EM radiation travels at c. Energy increases with frequency: **gamma rays carry far more energy per photon than radio waves**.

Check Your Understanding

Rank these from **longest** to **shortest** wavelength:

(a) Green light (visible)
(b) X-rays used in medical imaging
(c) Microwaves used in a microwave oven
(d) UV light causing sunburn

Come to lecture to see the answer.

Blackbody Radiation and Planck's Quantization

Blackbody Radiation Paradox: The Ultraviolet Catastrophe

- Theories in classical physics (physics before quantum mechanics was developed) predicted that a room-temperature object should emit **infinite** UV energy.
- This is obviously incorrect and did not match experimental results.
- In 1900, Max Planck developed a new theory that resolved this "catastrophe".

Planck's Theory of Quantized Energy

Energy is not continuous, but **quantized**. Atoms can only emit or absorb energy in discrete packets called **quanta** (photons):

$$\boxed{E = h\nu = \frac{hc}{\lambda}}$$

- $h = 6.626 \times 10^{-34}\,\mathrm{J\,s}$ (Planck's constant)
- ν = frequency of the light emitted/absorbed

Example: Photon Energy

Neon Sign

A neon sign emits orange-red light at $\lambda = 640\,\text{nm}$. What is the energy of one photon?

Given:

$$\lambda = 640\,\text{nm} = 6.40 \times 10^{-7}\,\text{m}$$
$$h = 6.626 \times 10^{-34}\,\text{J s}$$
$$c = 2.998 \times 10^{8}\,\frac{\text{m}}{\text{s}}$$

$$E = \frac{hc}{\lambda}$$
$$= \frac{(6.626 \times 10^{-34}\,\text{J s})(2.998 \times 10^{8}\,\frac{\text{m}}{\text{s}})}{6.40 \times 10^{-7}\,\text{m}}$$

$$\boxed{E = 3.10 \times 10^{-19}\,\text{J}}$$

The Photoelectric Effect

Observation

Shining light on a metal surface ejects electrons if the light exceeds a minimum frequency ν, regardless of intensity I.

Classical Prediction (Incorrect)

Any light, if bright enough, should eject electrons.

Einstein's Explanation (1905)

Light comes in **photons**. Each photon carries $E = h\nu$. Only a photon with $E \geq$ binding energy of the electron can eject it.

- $I \propto N_{photons}$
- $\nu \propto KE_{electrons}$

Key Points

- Increasing **intensity** $\Rightarrow$ more photons $\Rightarrow$ more electrons ejected
- Increasing **frequency** $\Rightarrow$ each photon has more energy $\Rightarrow$ electrons ejected with greater kinetic energy
- Below threshold frequency: **no** electrons ejected, ever

Check Your Understanding

A metal has a photoelectric threshold of 4.0×10^{14} Hz.

You shine bright red light (frequency $= 4.5 \times 10^{14}$ Hz) on it.

Then you replace it with dim blue light (frequency $= 7.0 \times 10^{14}$ Hz).

Which light ejects electrons? Does brightness matter here?

Come to lecture to see the answer.

Atomic Line Spectra

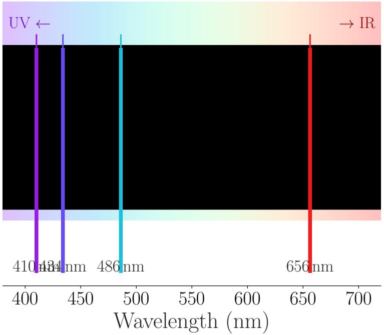
Continuous vs. Line Spectra

- **Continuous spectrum**: all wavelengths present (e.g., sunlight through a prism ⇒ rainbow)
- **Line (emission) spectrum**: only discrete wavelengths, unique to each element

Hydrogen Emission Series

- **Lyman**: UV region ($n_f = 1$)
- **Balmer**: Visible region ($n_f = 2$) — 4 lines: 656, 486, 434, 410 nm
- **Paschen**: IR region ($n_f = 3$)

Hydrogen Emission Spectrum (Balmer Seri



The Rydberg Formula

Empirical Formula for Hydrogen

Johann Balmer and Johannes Rydberg found an empirical formula predicting hydrogen's emission wavelengths:

$$\boxed{\frac{1}{\lambda} = R_\infty \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)}$$

- $R_\infty = 1.097 \times 10^7 \, \frac{1}{\mathrm{m}}$ (Rydberg constant)
- n_1, n_2 are positive integers with $n_2 > n_1$
- $n_1 = 1$: Lyman series (UV); $n_1 = 2$: Balmer (visible); $n_1 = 3$: Paschen (IR)

The formula was empirical — no one knew *why* it worked until Bohr in 1913.

Part II

The Bohr Model

Outline of Part II: The Bohr Model

The Bohr Model

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The Bohr Model

The Nuclear Atom's Problem

Rutherford's Model (1911)

A tiny, dense nucleus with electrons orbiting around it — like planets around the sun.

The Fatal Flaw

Classical electromagnetism: an accelerating charge **radiates energy**. Orbiting electrons must continuously lose energy $\Rightarrow$ spiral into the nucleus in $\sim 10^{-11}$ s. Atoms should not exist!

Yet atoms are stable, and emit only specific wavelengths. Something is fundamentally different at the atomic scale.

Bohr's Postulates (1913)

Niels Bohr's Key Assumptions

1. Electrons orbit the nucleus in **fixed, quantized orbits** ("shells") with specific energies. While in these orbits, they do **not** radiate energy.
2. An electron can only gain or lose energy by **jumping between orbits**:

$$\boxed{|\Delta E| = |E_f - E_i| = h\nu}$$

3. The orbit's energy is determined by the **principal quantum number** n $(n = 1, 2, 3, \ldots)$.

$n = 1$: **ground state** (lowest energy, closest to nucleus)
$n > 1$: **excited states** (higher energy, farther from nucleus)

Bohr Energy Levels for Hydrogen

Quantized Energies

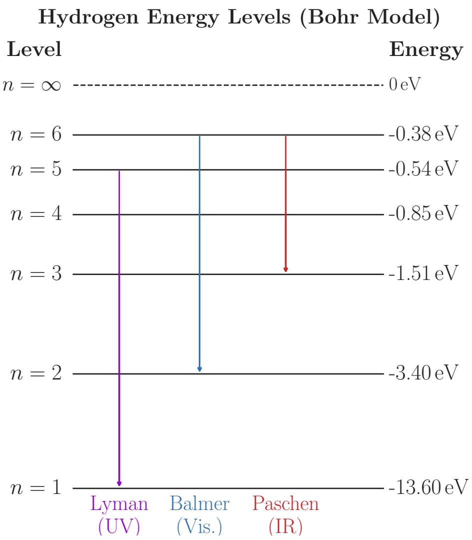
For hydrogen ($Z = 1$):

$$\boxed{E_n = -\frac{k}{n^2}}$$

- $k = 2.179 \times 10^{-18}\,\mathrm{J} = 13.6\,\mathrm{eV}$
- Negative sign: bound electron has lower energy than a free electron (E defined as $E = 0$ at $n = \infty$)
- As $n \to \infty$: $E_n \to 0$ (ionization)

Energy of Bound Electron in 2nd Energy Level

$$E_2 = -\frac{13.6\,\mathrm{eV}}{2^2} = -3.40\,\mathrm{eV}$$



Example: Energy of an Emission

Hydrogen Transition

An electron in hydrogen drops from $n = 4$ to $n = 2$. Find the energy and wavelength of the emitted photon.

$$\Delta E = -k\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$= -2.179 \times 10^{-18}\,\mathrm{J}\left(\frac{1}{4} - \frac{1}{16}\right)$$

$$\boxed{\Delta E = -4.086 \times 10^{-19}\,\mathrm{J}}$$

$$\lambda = \frac{hc}{|\Delta E|}$$

$$= \frac{(6.626 \times 10^{-34}\,\mathrm{J\,s})(2.998 \times 10^{8}\,\frac{\mathrm{m}}{\mathrm{s}})}{4.086 \times 10^{-19}\,\mathrm{J}}$$

$$\boxed{\lambda = 486\,\mathrm{nm}}$$

Blue-green visible light (Balmer series)

Check Your Understanding

An electron in hydrogen transitions from $n = 3$ to $n = 1$.

Is this a **Lyman**, **Balmer**, or **Paschen** series line?

Is the photon **emitted** or **absorbed**? Is the energy of the atom increasing or decreasing?

Come to lecture to see the answer.

Limitations of the Bohr Model

Successes

- Correctly predicted the **line spectrum of hydrogen**
- Bohr's theoretical Rydberg constant matched experiment
- Introduced the concept of quantized energy levels

Failures

- Failed for **multi-electron atoms** (even helium)
- Could not explain **fine structure** (closely spaced spectral lines)
- Electrons do not travel in well-defined circular orbits
- Mixed classical and quantum ideas in an inconsistent way

The Bohr model was a stepping stone. A deeper theory was needed.

Part III

Development of Quantum Theory

Outline of Part III: Development of Quantum Theory

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Development of Quantum Theory

Wave-Particle Duality of Matter

de Broglie's Hypothesis (1924)

If light (a wave) can behave like a particle, perhaps particles (electrons, protons) can behave like waves. de Broglie proposed:

$$\boxed{\lambda = \frac{h}{mv} = \frac{h}{p}}$$

- m = mass, v = speed, $p = mv$ = momentum
- This wavelength is only noticeable for very small (atomic-scale) objects

Electron Wavelength

An electron ($m = 9.11 \times 10^{-31}$ kg) moving at $v = 1 \times 10^{7}\ \frac{\mathrm{m}}{\mathrm{s}}$:

$$\lambda = \frac{6.626 \times 10^{-34}\ \mathrm{J\,s}}{9.11 \times 10^{-31}\ \mathrm{kg}\,(1 \times 10^{7}\ \frac{\mathrm{m}}{\mathrm{s}})}$$

$$\boxed{\lambda = 7 \times 10^{-11}\ \mathrm{m}} \qquad \approx \text{atomic scale!}$$

When is Quantum Mechanics Needed?

Quantum mechanics is needed to describe an object when

$$\lambda > L$$

where L represents the length scale of the object. This could be the width, the height, length, diameter, etc.

Check Your Understanding

A baseball (mass $= 0.14\,\mathrm{kg}$) travels at $40\,\frac{\mathrm{m}}{\mathrm{s}}$.

Would you expect its de Broglie wavelength to be observable?

(You don't need to calculate — reason from the formula.)

Come to lecture to see the answer.

The Heisenberg Uncertainty Principle

It is **fundamentally impossible** to know simultaneously and precisely both the position and the momentum of a particle:

$$\boxed{\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2}} \qquad \left(\hbar = \frac{h}{2\pi}\right)$$

- Δx = uncertainty in position
- Δp_x = uncertainty in momentum

- This is **not** a limitation of our instruments; it is a property of nature.
- Electrons do **not** travel in neat orbits — their positions are described by **probability distributions**.
- The Bohr orbit picture is fundamentally incorrect.

The Schrödinger Equation and Wave Functions

Quantum Mechanical Description

Erwin Schrödinger (1926) described electrons as **three-dimensional standing waves**:

$$\hat{H}\psi = E\psi$$

- ψ (psi) = **wave function** (atomic orbital): a mathematical description of an electron's state
- $\hat{H}$ = Hamiltonian operator (total energy operator)
- Solutions give allowed energies E and orbital shapes ψ

Born Interpretation

$|\psi|^2$ = **probability density** — the probability of finding an electron at a given point in space.

An **orbital** is a region of space where an electron is most likely to be found — **not** a circular path.

Quantum Numbers: Overview

Each electron in an atom is described by **four quantum numbers**:

Name	Symbol	Allowed Values	Determines
Principal	n	$1, 2, 3, \ldots$	Energy level (shell)
Angular momentum	ℓ	0 to $n-1$	Orbital shape (subshell)
Magnetic	m_ℓ	$-\ell$ to $+\ell$	Orbital orientation
Spin	m_s	$+\frac{1}{2}, -\frac{1}{2}$	Electron spin

Every electron in an atom has a **unique set** of all four quantum numbers (Pauli Exclusion Principle).

Principal Quantum Number n

- $n = 1, 2, 3, 4, \ldots$ (positive integers only)
- Defines the **shell** (energy level)
- As n increases: orbital is **larger** and **higher in energy**
- Maximum electrons in shell n: $2n^2$

n	Shell	Max electrons
1	K	2
2	L	8
3	M	18
4	N	32

n is like a "floor" in a building.

Higher floor = more energy and more space, but farther from the "ground."

Angular Momentum Quantum Number ℓ

- $\ell = 0, 1, 2, \ldots, n-1$ (depends on n)
- Defines the **subshell** and orbital **shape**

ℓ	**Subshell**	**Shape**	**Orbitals** $(2\ell + 1)$
0	s	Spherical	1
1	p	Dumbbell (lobe)	3
2	d	Cloverleaf	5
3	f	Complex	7

For $n = 3$: ℓ can be 0, 1, or 2, giving subshells 3s, 3p, 3d.

Magnetic and Spin Quantum Numbers

Magnetic Quantum Number m_ℓ

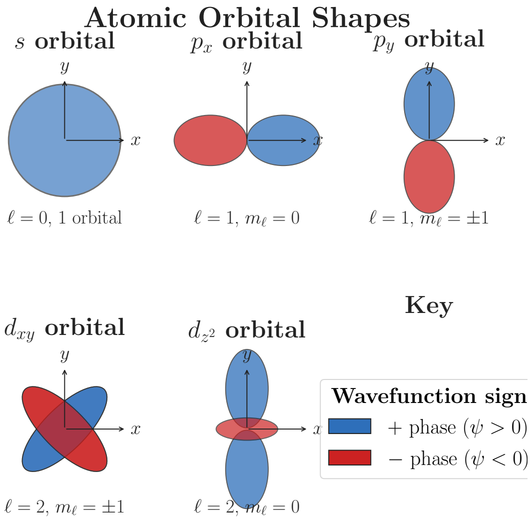
- $m_\ell = -\ell, \ldots, 0, \ldots, +\ell$
- Specifies **orientation** of the orbital in space
- For p ($\ell = 1$): $m_\ell = -1, 0, +1 \Rightarrow$ three p orbitals: p_x, p_y, p_z

Spin Quantum Number m_s

- $m_s = +\frac{1}{2}$ ("spin up", $\uparrow$) or $-\frac{1}{2}$ ("spin down", $\downarrow$)
- Electrons behave as tiny magnets; spin gives a magnetic dipole moment
- Each orbital holds **at most 2 electrons** with opposite spins

Pauli Exclusion Principle: No two electrons in an atom can have the same set of all four quantum numbers. This limits each orbital to 2 electrons.

Orbital Shapes



- **s**: 1 spherical orbital; no angular nodes
- **p**: 3 dumbbell orbitals along x, y, z; 1 angular node
- **d**: 5 orbitals, more complex; 2 angular nodes
- **f**: 7 orbitals; 3 angular nodes

Nodes = regions of zero probability. More nodes = higher energy within a shell.

Check Your Understanding

For $n = 3$:

(a) What are the allowed values of ℓ?

(b) How many orbitals are in the 3d subshell?

(c) What is the maximum number of electrons in the $n = 3$ shell?

Come to lecture to see the answer.

Part IV

Electron Configurations

Outline of Part IV: Electron Configurations

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Orbital Energies and the Aufbau Principle

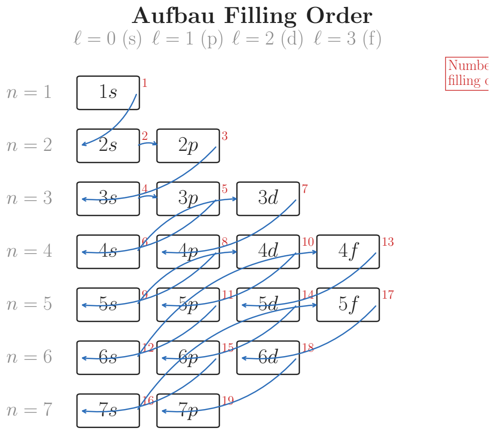
Aufbau Principle

"Build up" the electron configuration by filling orbitals from **lowest to highest energy**:

$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d \cdots$$

Note: 4s fills *before* 3d!

The order follows from the $n+\ell$ rule: lower $n+\ell$ fills first; when equal, lower n fills first.



Hund's Rule

Hund's Rule of Maximum Multiplicity

Within a subshell (set of degenerate orbitals), electrons fill each orbital **singly** before pairing up, and all unpaired electrons have the **same spin**.

Nitrogen ($Z = 7$): $1s^2\ 2s^2\ 2p^3$

$$\underset{1s}{\boxed{\uparrow\downarrow}} \quad \underset{2s}{\boxed{\uparrow\downarrow}} \quad \underset{2p}{\boxed{\uparrow}\boxed{\uparrow}\boxed{\uparrow}}$$

3 unpaired electrons — one in each 2p orbital

The half-filled $2p^3$ configuration of nitrogen is especially **stable** due to exchange energy.

Writing Electron Configurations

Notation: $n\ell^x$

- n = principal quantum number, ℓ = subshell letter, x = number of electrons in that subshell
- Example: $\mathbf{2p}^4$ = 4 electrons in the 2p subshell

Examples

Element	Z	**Configuration**
Hydrogen	1	$1s^1$
Helium	2	$1s^2$
Lithium	3	$1s^2\ 2s^1$
Carbon	6	$1s^2\ 2s^2\ 2p^2$
Nitrogen	7	$1s^2\ 2s^2\ 2p^3$
Oxygen	8	$1s^2\ 2s^2\ 2p^4$
Neon	10	$1s^2\ 2s^2\ 2p^6$
Sodium	11	$1s^2\ 2s^2\ 2p^6\ 3s^1$

Core and Valence Electrons

Valence Electrons

Electrons in the **outermost shell** (highest n).

- Determine **chemical reactivity**
- Form bonds with other atoms
- For main-group elements: the s and p electrons of the highest shell

Core Electrons

All **inner-shell** electrons not in the valence shell.

- Shield valence electrons from the full nuclear charge
- Relatively unreactive
- Abbreviated using a noble gas symbol: **[Ne]**, **[Ar]**, **[Kr]**, etc.

Sodium (Na, $Z = 11$): $1s^2\ 2s^2\ 2p^6\ 3s^1 = \mathbf{[Ne]\ 3s}^1$
Valence electrons: 1 Core electrons: 10

Check Your Understanding

Write the full electron configuration for **phosphorus** ($Z = 15$).

How many **valence electrons** does it have?

Write the abbreviated (noble gas) configuration.

Come to lecture to see the answer.

Classification by Electron Configuration

Main-Group Elements

Last electron enters **s or p** subshell.

Groups 1–2 (s-block) and 13–18 (p-block).

Transition Metals

Last electron enters a **d** subshell.

Groups 3–12. Variable oxidation states.

Inner Transition

Last electron enters an **f** subshell.

Lanthanides (4*f*) and actinides (5*f*).

Examples

- Fe ($Z = 26$): [Ar] $3d^6\ 4s^2$ (transition metal)
- La ($Z = 57$): [Xe] $5d^1\ 6s^2$ (lanthanide series begins)
- Cl ($Z = 17$): [Ne] $3s^2\ 3p^5$ (main-group, p-block)

Notable Exceptions: Cr and Cu

Chromium ($Z = 24$)

Expected: $[\mathrm{Ar}]\ 3\mathrm{d}^4\ 4\mathrm{s}^2$
Actual: $\mathbf{[Ar]\ 3d^5\ 4s^1}$

A **half-filled** 3d subshell ($3\mathrm{d}^5$) is especially stable. One electron shifts from 4s to 3d.

Copper ($Z = 29$)

Expected: $[\mathrm{Ar}]\ 3\mathrm{d}^9\ 4\mathrm{s}^2$
Actual: $\mathbf{[Ar]\ 3d^{10}\ 4s^1}$

A **completely filled** 3d subshell ($3\mathrm{d}^{10}$) is especially stable. One electron shifts from 4s to 3d.

Rule of thumb: half-filled and fully filled d and f subshells have extra stability due to exchange energy.

Electron Configurations of Ions

Forming Cations (remove electrons)

Remove electrons from the **highest-n subshell first**.

For transition metals, 4s electrons are removed before 3d!

Examples

- Na^+ ($Z = 11$, $-1e^-$): $1s^2\ 2s^2\ 2p^6$ = **[Ne]**
- Fe^{2+} ($Z = 26$, $-2e^-$): [Ar] $3d^6$ (removes $4s^2$ first)
- Fe^{3+} ($Z = 26$, $-3e^-$): [Ar] $3d^5$

Forming Anions (add electrons)

Add electrons following Aufbau principle.

Examples

- Cl^- ($Z = 17$, $+1e^-$): [Ne] $3s^2\ 3p^6$ = **[Ar]**
- O^{2-} ($Z = 8$, $+2e^-$): $1s^2\ 2s^2\ 2p^6$ = **[Ne]**

Check Your Understanding

Write the abbreviated electron configuration for:

(a) Calcium ($Z = 20$)

(b) Ca^{2+}

(c) Sulfur ($Z = 16$)

(d) S^{2-}

Come to lecture to see the answer.

Part V

Periodic Trends

Outline of Part V: Periodic Trends

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Effective Nuclear Charge

Definition

The **effective nuclear charge** Z_{eff} is the net positive charge "felt" by a valence electron after accounting for shielding by inner (core) electrons:

$$\boxed{Z_{\text{eff}} = Z - \sigma}$$

where σ = shielding constant (approximately equal to the number of core electrons).

Across Period 3

Element	Na	Mg	Al	Si
Z	11	12	13	14
Core e^-	10	10	10	10
Z_{eff}	≈ 1	≈ 2	≈ 3	≈ 4

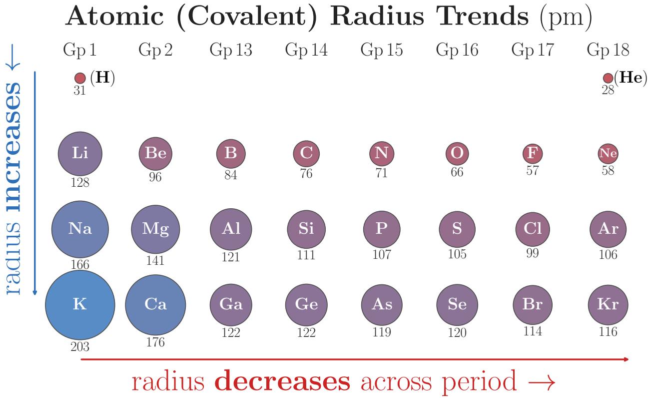
Atomic (Covalent) Radius

Definition

Covalent radius: half the distance between the nuclei of two identical atoms bonded together.

Trends

- **Across a period (left → right)**: radius **decreases** — Z_{eff} increases, pulling electrons closer
- **Down a group**: radius **increases** — adding a new electron shell farther from the nucleus



Largest atom: Cs (bottom-left).
Smallest non-noble-gas atom: F (top-right).

Ionic Radii

Cations (positive ions)

- **Smaller** than the parent atom
- Removing electrons reduces electron–electron repulsion
- Z_{eff} per remaining electron **increases**

Na: 186 pm Na^+: 102 pm

Anions (negative ions)

- **Larger** than the parent atom
- Adding electrons increases repulsion
- Z_{eff} per electron **decreases**

Cl: 99 pm Cl^-: 181 pm

Isoelectronic series: same number of electrons; nuclear charge determines size. *e.g.* O^{2-}, F^-, *Ne*, Na^+, Mg^{2+} *all have 10 electrons — radius decreases with increasing Z.*

Check Your Understanding

Arrange these atoms in order of **increasing** atomic radius (smallest to largest):

Na Cl K Br

Come to lecture to see the answer.

Ionization Energy

First Ionization Energy (IE_1)

Energy required to remove the **most loosely bound electron** from a gaseous atom:

$$X(g) \longrightarrow X^+(g) + e^- \qquad \Delta E = IE_1 > 0$$

Trends

- **Across a period**: IE generally **increases** (higher Z_{eff} holds electrons more tightly)
- **Down a group**: IE **decreases** (electrons farther from nucleus, more shielded)

Notable Deviations

- $IE_1(B) < IE_1(Be)$: B's 2p electron is easier to remove than Be's 2s
- $IE_1(O) < IE_1(N)$: O must pair a 2p electron, increasing repulsion

Successive Ionization Energies

Each successive electron removal requires **more energy**. When a **core electron** is reached, the ionization energy **jumps dramatically**.

Sodium (Na): Successive IE values (kJ/mol)

IE	IE_1	IE_2	IE_3	IE_4	IE_5	IE_6
kJ/mol	496	4562	6912	9543	13 353	16 610

The **enormous jump** from IE_1 to IE_2 for Na reveals that Na has **1 valence electron**. This confirms the electron configuration and explains why Na forms Na^+ but not Na^{2+}.

Electron Affinity

Definition

The **electron affinity** (EA) is the energy change when a gaseous atom *gains* an electron:

$$X(g) + e^- \longrightarrow X^-(g) \qquad \Delta E = \text{EA}$$

- **Negative EA**: energy is *released* (exothermic); element attracts electrons readily (e.g., Cl: EA $= -349\,\frac{\text{kJ}}{\text{mol}}$)
- **Positive EA**: energy must be *added*; element resists gaining electrons (noble gases)

Trends

- Generally becomes **more negative** across a period (left $\rightarrow$ right)
- Halogens have the most negative EA (Cl $>$ F: F's small size causes high electron repulsion)
- Groups 2, 15, and 18 show **deviations** due to filled or half-filled subshells

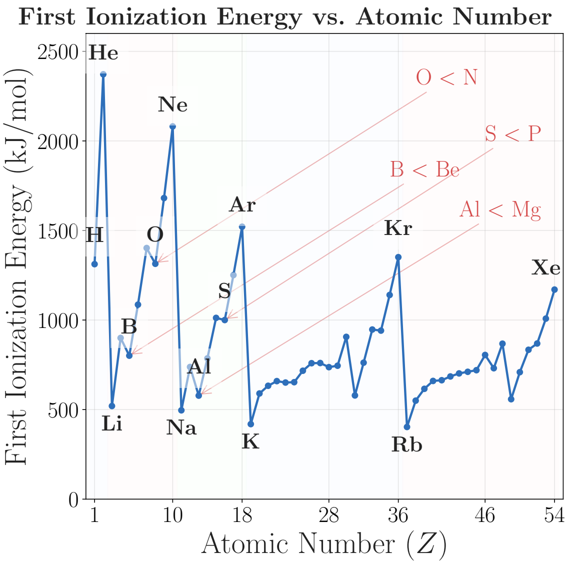
Check Your Understanding

Which has the **higher** first ionization energy — Na or Cl? Why?

Which has the **more negative** electron affinity — F or Ne? Why?

Come to lecture to see the answer.

Periodic Trends in Ionization Energy



Key Points

- Noble gases: highest IE (complete shells)
- Alkali metals: lowest IE (1 valence electron)
- General increase across each period
- Notable dips at Group 13 and Group 16

The "sawtooth" pattern across periods reflects subshell structure.

Summary of Periodic Trends

Across a Period (left → right)

- Atomic radius **decreases**
- Ionization energy **increases** (generally)
- Electron affinity **more negative** (generally)
- Z_{eff} **increases**

Down a Group

- Atomic radius **increases**
- Ionization energy **decreases**
- Electron affinity **less negative** (generally)
- More shielding from core electrons

Root cause: All trends trace back to **effective nuclear charge** (Z_{eff}) and the **principal quantum number** (n) of the valence electrons.

Check Your Understanding

Arrange the following elements in order of **increasing first ionization energy**:

Mg Al Na S

(All are Period 3 elements — use trends, not memorized values.)

Come to lecture to see the answer.

Outline of Part V: Periodic Trends

Key Equations — Chapter 6

Equation	Meaning
$c = \lambda\nu$	Speed of light: wavelength $\times$ frequency
$E = h\nu = hc/\lambda$	Photon energy (Planck, $h = 6.626 \times 10^{-34}\,\mathrm{J\,s}$)
$\frac{1}{\lambda} = R_\infty\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$	Rydberg formula (H emission wavelengths)
$E_n = -k/n^2$	Bohr energy levels ($k = 2.179 \times 10^{-18}\,\mathrm{J}$)
$\lambda = h/(mv)$	de Broglie wavelength of a particle
$\Delta x \cdot \Delta p \geq \hbar/2$	Heisenberg Uncertainty Principle

Chapter 6 Summary

- **EM radiation**: $c = \lambda\nu$; energy quantized as $E = h\nu$; light has wave *and* particle character.
- **Photoelectric effect**: confirms photon picture; threshold frequency required regardless of intensity.
- **Bohr model**: quantized orbits explain hydrogen's line spectrum; $E_n = -k/n^2$.
- **Wave-particle duality**: electrons have de Broglie wavelength $\lambda = h/mv$; Heisenberg uncertainty forbids precise orbits.
- **Quantum numbers**: n (size/energy), ℓ (shape), m_ℓ (orientation), m_s (spin) uniquely specify each electron.
- **Electron configurations**: Aufbau principle, Hund's rule, and Pauli Exclusion Principle govern filling.
- **Periodic trends**: atomic radius, IE, and EA all controlled by Z_{eff} and shell number.